

Geodesic Trajectory Generation and Differential-Flatness Control of a PMSM

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Abstract

This memo develops a geometric control architecture for a permanent-magnet synchronous motor (PMSM).

The main idea is to combine two concepts:

1. geodesic trajectory generation on the PMSM current manifold, and
2. differential-flatness based feedforward control.

The magnetic co-energy of the machine is used to define a Riemannian metric on the (i_d, i_q) current space. Geodesics of this metric provide intrinsic current trajectories.

Differential flatness is then used to transform a desired current trajectory and its derivatives into the corresponding stator voltage commands (v_d, v_q) .

The resulting architecture separates trajectory generation from control:

Magnetic geometry $\rightarrow$ Geodesic trajectory $\rightarrow$ Flatness transformation $\rightarrow$ Voltage command

This provides a possible foundation for a model-based PMSM control architecture suitable for real-time embedded implementation.

1 Introduction

Conventional PMSM control commonly separates trajectory generation from current regulation. A reference such as

$$i_d^* = 0$$

and a torque-producing i_q^* is supplied to a current controller, which then determines the required stator voltage.

The approach considered here is different.

The current state

$$x = \begin{bmatrix} i_d \\ i_q \end{bmatrix}$$

is regarded as a point on a state manifold. The magnetic properties of the PMSM define a metric on this space. A desired transition between two operating points can therefore be formulated as a geodesic problem.

Once a desired trajectory

$$x^*(t) = \begin{bmatrix} i_d^*(t) \\ i_q^*(t) \end{bmatrix}$$

has been obtained, differential flatness provides a direct transformation from the trajectory to the required control voltage.

Thus the proposed architecture is

$$\boxed{\text{Geometric trajectory generation} + \text{Differential-flatness control}}$$

rather than treating these as competing control methods.

2 PMSM Electrical Model

Consider the PMSM model in the rotating dq reference frame.

The electrical dynamics are

$$\dot{i}_d = -\frac{R_s}{L_d}i_d + \frac{L_q}{L_d}\omega_e i_q + \frac{1}{L_d}v_d, \quad (1)$$

$$\dot{i}_q = -\frac{R_s}{L_q}i_q - \frac{L_d}{L_q}\omega_e i_d - \frac{\psi_m}{L_q}\omega_e + \frac{1}{L_q}v_q. \quad (2)$$

Here

- i_d, i_q are the dq currents,
- v_d, v_q are the dq voltages,
- R_s is the stator resistance,
- L_d, L_q are the dq inductances,
- ψ_m is the permanent-magnet flux linkage,
- ω_e is the electrical angular velocity.

The electromagnetic torque is

$$T_e = \frac{3}{2}p [\psi_m i_q + (L_d - L_q)i_d i_q], \quad (3)$$

where p denotes the number of pole pairs.

3 State-Space Form

The electrical model can be written as

$$\dot{x} = f(x, \omega_e) + Bu \quad (4)$$

with

$$x = \begin{bmatrix} i_d \\ i_q \end{bmatrix}, \quad u = \begin{bmatrix} v_d \\ v_q \end{bmatrix}. \quad (5)$$

For the standard constant-parameter model,

$$B = \begin{bmatrix} 1/L_d & 0 \\ 0 & 1/L_q \end{bmatrix}. \quad (6)$$

The control input therefore appears directly in the first derivative of the current state.

4 Differential Flatness

Choose the electrical currents as the output:

$$y = \begin{bmatrix} i_d \\ i_q \end{bmatrix}. \quad (7)$$

Since the inputs appear directly in the first derivatives of the outputs, the PMSM electrical subsystem can be inverted with respect to the voltage inputs.

Solving the electrical equations for v_d and v_q gives

$$\boxed{v_d = L_d \dot{i}_d + R_s i_d - L_q \omega_e i_q} \quad (8)$$

and

$$\boxed{v_q = L_q \dot{i}_q + R_s i_q + L_d \omega_e i_d + \psi_m \omega_e} \quad (9)$$

Therefore, for a desired trajectory

$$i_d^*(t), \quad i_q^*(t), \quad (10)$$

the corresponding feedforward voltage is

$$\boxed{v_d^* = L_d \dot{i}_d^* + R_s i_d^* - L_q \omega_e i_q^*} \quad (11)$$

and

$$\boxed{v_q^* = L_q \dot{i}_q^* + R_s i_q^* + L_d \omega_e i_d^* + \psi_m \omega_e}. \quad (12)$$

This is the differential-flatness part of the controller.

5 Magnetic Co-Energy

The trajectory generator can be constructed from the magnetic co-energy

$$W_c = W_c(i_d, i_q). \quad (13)$$

For a linear PMSM,

$$W_c = \frac{1}{2} L_d i_d^2 + \frac{1}{2} L_q i_q^2 + \psi_m i_q. \quad (14)$$

The Hessian of the co-energy is

$$g_{ij} = \frac{\partial^2 W_c}{\partial x^i \partial x^j}. \quad (15)$$

Thus,

$$g = \nabla^2 W_c. \quad (16)$$

For the linear model,

$$g = \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix}. \quad (17)$$

The metric is therefore constant.

6 Nonlinear Magnetic Model

For a real machine, magnetic saturation and cross-saturation can make the co-energy nonlinear:

$$W_c = W_c(i_d, i_q). \quad (18)$$

The resulting metric is

$$g(i_d, i_q) = \nabla^2 W_c(i_d, i_q). \quad (19)$$

In this case the metric varies over the current space.

The inverse metric is denoted by

$$g^{ij} = (g_{ij})^{-1}. \quad (20)$$

The corresponding Christoffel symbols are

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} \left(\frac{\partial g_{j\ell}}{\partial x^i} + \frac{\partial g_{i\ell}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^\ell} \right). \quad (21)$$

7 Geodesic Trajectory

A geodesic satisfies

$$\ddot{x}^k + \Gamma_{ij}^k(x) \dot{x}^i \dot{x}^j = 0. \quad (22)$$

Equivalently,

$$\dot{x}^k = v^k, \quad (23)$$

$$\dot{v}^k = -\Gamma_{ij}^k(x) v^i v^j. \quad (24)$$

The solution

$$x^*(t) = \begin{bmatrix} i_d^*(t) \\ i_q^*(t) \end{bmatrix} \quad (25)$$

provides the desired current trajectory.

8 Important Special Case

For the linear constant-parameter PMSM,

$$\frac{\partial g_{ij}}{\partial x^k} = 0. \quad (26)$$

Consequently,

$$\Gamma_{ij}^k = 0. \quad (27)$$

The geodesic equation becomes

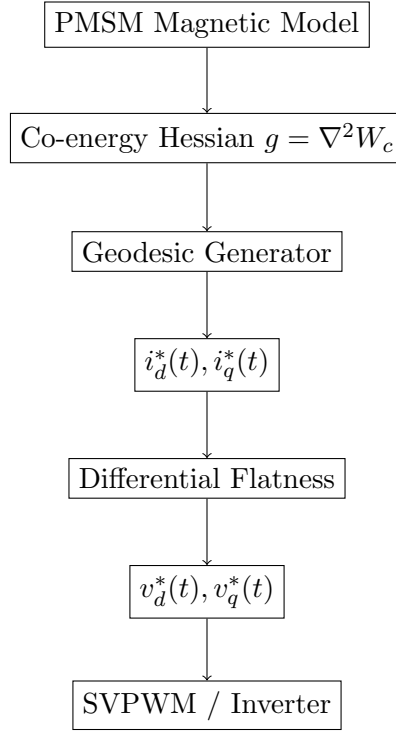
$$\ddot{x} = 0. \quad (28)$$

Hence the geodesics are straight lines in the (i_d, i_q) plane.

This observation is important: a nontrivial geometric trajectory requires a nonlinear magnetic model, for example one including saturation and cross-saturation.

9 Combined Architecture

The proposed control architecture is



10 Feedback Correction

The differential-flatness equations provide a model-based feedforward command. In practice, model uncertainty, parameter variation, sampling effects, inverter nonlinearities, and disturbances require feedback correction.

Therefore,

$$u = u_{\text{ff}} + u_{\text{fb}}, \quad (29)$$

where

$$u_{\text{ff}} = \begin{bmatrix} v_d^* \\ v_q^* \end{bmatrix}. \quad (30)$$

The feedback term can be kept relatively small if the model and trajectory are sufficiently accurate.

11 Embedded Implementation

The intended implementation separates the mathematical layers:

1. magnetic model,
2. metric calculation,
3. geodesic trajectory generation,
4. differential-flatness transformation,

5. feedback correction,
6. voltage limitation,
7. inverse Park transformation,
8. SVPWM.

The resulting implementation can therefore be represented as

$$\boxed{W_c \rightarrow g \rightarrow \Gamma \rightarrow x^* \rightarrow \dot{x}^* \rightarrow u^* \rightarrow \text{SVPWM}.} \quad (31)$$

This structure is particularly suitable for a model-based embedded control framework in which the mathematical controller is developed independently of the target microcontroller.

12 Relation to GALEC

The proposed architecture can also be expressed naturally in GALEC.

A GALEC controller can describe the mathematical relationship between the desired flat output and the control input without exposing the implementation details of the target microcontroller.

Conceptually,

```
MagneticModel
-> Metric
-> Geodesic
-> FlatOutput
-> FlatnessControl
-> Voltage
-> SVPWM
```

The GALEC compiler can subsequently transform these operations into restricted, statically analyzable embedded C code.

This creates a separation between

$$\text{control mathematics} \quad (32)$$

and

$$\text{microcontroller implementation.} \quad (33)$$

13 Conclusion

The combination of geodesic trajectory generation and differential flatness provides a coherent geometric formulation of PMSM control.

Differential flatness answers the question

Given a desired current trajectory, what voltage is required to produce it?

Geometric trajectory generation answers the complementary question

Which current trajectory should be followed between two operating points?

The proposed architecture therefore combines

geometry for trajectory generation

with

differential flatness for control inversion.

For a nonlinear magnetic model, the geometry contains information about the actual magnetic structure of the machine. This provides a possible foundation for further investigation of geometry-aware PMSM trajectory generation and embedded control.